Real Analysis II: Study Note

Chapter 2: Integration Theory

Tianqi Zhang

Department of Economics

University of Southern California

Spring 2026

Contents

1 Integration of Simple Functions **3**

2 Integration of Bounded Functions **4**

3 Integration of Positive Functions **5**

4 The Full Lebesgue Integral **5**

5 Convergence Theorems **6**

5.1 Common Counterexamples 7

6 Convergence in Measure **7**

7 Properties of Integrability **8**

Proofs of Key Results **9**

1 Integration of Simple Functions

The canonical failure of Riemann integration is $\chi_{\mathbb{Q}} : [0,1] \to \mathbb{R}$: dense irrationals force every upper sum to 1 and every lower sum to 0, so the two never meet.

Roadmap: BCT → Fatou → MCT → DCT

The central question throughout is: *when can we pass a limit through an integral?* Each theorem relaxes the price of admission. BCT imposes the most rigid constraints — a uniform bound M on the functions and a finite-measure domain, essentially trapping everything in a box. Fatou's Lemma then allows the functions to be unbounded, at the cost of only an inequality. MCT trades the uniform bound for monotonicity, letting the functions grow so long as they do so in an orderly fashion. DCT is the culmination: we need only a single integrable dominating function, making it the workhorse of the subject.

Definition 1.1: Integral of a Simple Function

Given a simple function $\varphi = \sum_{i=1}^{k} c_i \chi_{E_i}$, define

$$\int \varphi \;:=\; \sum_i c_i\, m(E_i).$$

More generally, against a measure μ: $\int \varphi \, \mathrm{d}\mu$; with explicit variable: $\int \varphi(x) \, \mathrm{d}\mu(x)$; over a sub-domain E:

$$\int_E \varphi \;:=\; \sum_i c_i\, m(E_i \cap E).$$

Proposition 1.2: Properties of the Simple Integral[1]

For simple functions:

(i) **Independence of representation.** If $\varphi = \sum_i c_i \chi_{E_i} = \sum_j b_j \chi_{F_j}$, then $\sum_i c_i m(E_i) = \sum_j b_j m(F_j)$.

(ii) **Monotonicity.** $\varphi_1 \le \varphi_2 \Rightarrow \int \varphi_1 \le \int \varphi_2$.

(iii) **Linearity.** $\int (a_1\varphi_1 + a_2\varphi_2) = a_1 \int \varphi_1 + a_2 \int \varphi_2$.

(iv) **Triangle inequality.** $\left|\int \varphi\right| \le \int |\varphi|$.

[1] Property (i) is non-trivial and must be verified: the integral must not depend on the particular way φ is written as a linear combination of indicators.

2 Integration of Bounded Functions

Strategy: (i) approximate f by simple functions; (ii) use Egorov's theorem to ensure convergence of integrals.

Lemma 2.1

Let f be bounded on E with $m(E) < \infty$, and let $\{\varphi_j\}_{j\in\mathbb{N}}$ be simple functions with $\varphi_j \to f$ on E. Then:

(i) If $|\varphi_j| \leq |f|$, then $\lim_{j\to\infty} \int_E \varphi_j$ exists.

(ii) If $f = 0$ a.e., then $\lim_{j\to\infty} \int_E \varphi_j = 0$.

Definition 2.2: Support

The **support** of f is $\operatorname{supp}(f) := \{f \neq 0\}$.

Definition 2.3: Lebesgue Integral — Bounded Case[2]

Given $f : \mathbb{R}^d \to \mathbb{R}$ bounded with $m(\operatorname{supp}(f)) < \infty$, define

$$\int f \;:=\; \lim_{n\to\infty} \int \varphi_n$$

for any sequence of simple functions $\{\varphi_n\}$ satisfying: $\varphi_n \to f$, $\operatorname{supp}(\varphi_n) \subset \operatorname{supp}(f)$, $|\varphi_n| \leq |f|$.

Example. $\chi_{\mathbb{Q}} : [0,1] \to \{0,1\}$ is *not* Riemann integrable (dense discontinuities everywhere). However, $\int_{[0,1]} \chi_{\mathbb{Q}} = 0$ in the Lebesgue sense, since $\mathbb{Q}$ has measure zero.

Theorem 2.4: Bounded Convergence Theorem[3]

Suppose $\{f_n\}_{n\in\mathbb{N}}$ satisfies $|f_n| \leq M$ and $\operatorname{supp}(f_n) \subset E$ for all n, with $m(E) < \infty$. If $f_n \to f$ a.e. on E, then

$$\lim_{n\to\infty} \int_E |f_n - f| = 0, \qquad \text{and in particular} \qquad \lim_{n\to\infty} \int f_n = \int f.$$

[2] Two facts must be checked: (i) the limit exists (by Lemma 2.1); (ii) $\int f$ is independent of the choice of approximating sequence $\varphi_n \to f$.

[3] Prototype for Lebesgue's DCT (Theorem 5.3). The BCT constrains f_n in a rectangular box $M \cdot \chi_E$. The DCT replaces this rigid box with any integrable "envelope" g.

3 Integration of Positive Functions

Definition 3.1: Lebesgue Integral — Positive Case

For $f : \mathbb{R}^d \to [0, \infty]$ (i.e., $f \geq 0$), define

$$\int f \;:=\; \sup\left\{ \int g \;:\; g \text{ bounded},\; m(\operatorname{supp}(g)) < \infty,\; 0 \leq g \leq f \right\}.$$

Proposition 3.2

The integral $\int f$ is monotone and linear over positive functions.

4 The Full Lebesgue Integral

Definition 4.1: Lebesgue Integral — General Case

Decompose any $f = f^+ - f^-$, where $f^+ := \max(f, 0)$ and $f^- := \max(-f, 0)$. Define

$$\int f \;:=\; \int f^+ - \int f^-,$$

provided both $\int f^+$ and $\int f^-$ are finite.

$f : E \to \mathbb{R}$ is **integrable** if $\displaystyle\int |f| < \infty$, equivalently $\int f^+ < \infty$ and $\int f^- < \infty$.

Extension to $\mathbb{C}$. For $f : \mathbb{R}^d \to \mathbb{C}$, write $f = f_1 - f_2 + if_3 - if_4$ with each $f_k \geq 0$. Say f is integrable if $\int |f| < \infty$.

Proposition 4.2

If f is integrable, then $\int f$ is linear and monotone.

Remark 4.1. *Nothing in Sections 1–4 is specific to Lebesgue measure; the construction works for any measure μ on any measurable space $(X, \mathcal{F}, \mu)$.*

Theorem 4.3: Chebyshev's Inequality

For $f \geq 0$ and $\alpha > 0$,

$$m(\{f \geq \alpha\}) \;\leq\; \frac{1}{\alpha} \int f.$$

Extension (Markov's inequality): $m(\{f \geq \alpha\}) = m(\{f^p \geq \alpha^p\}) \leq \dfrac{1}{\alpha^p} \displaystyle\int f^p$[a].

[a] More commonly seen as $P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$ $\forall a > 0$ in probability theory.

Two immediate corollaries:

Lemma 4.4

For $f : \mathbb{R}^d \to [0, \infty]$, if $\int f < \infty$, then $f < \infty$ a.e.

Lemma 4.5

For $f : \mathbb{R}^d \to [0, \infty]$, if $\int f = 0$, then $f = 0$ a.e.

5 Convergence Theorems

Central question: When does $f_n \to f$ imply $\int f_n \to \int f$, i.e. when can we exchange lim and $\int$?

Theorem 5.1: Fatou's Lemma

If $f_n \geq 0$ for all n and $f_n \to f$ a.e., then

$$\int f \;\leq\; \liminf_{n\to\infty} \int f_n.$$

Remark 5.1. *The inequality can be strict. See Counterexample (iii) below for a sequence where $\int f_n = 1$ for all n yet $\int f = 0$.*

Theorem 5.2: Monotone Convergence Theorem (MCT)

Let $f_k : \mathbb{R}^d \to [0, \infty]$ with $f_k \geq 0$. If $f_k \nearrow f$ monotonically (pointwise a.e.), then

$$\int f_k \;\longrightarrow\; \int f.$$

Corollary to MCT. If $u_n \geq 0$ for all n, then

$$\sum_{n=1}^{\infty} \int u_n \;=\; \int \sum_{n=1}^{\infty} u_n,$$

since the partial sums $\sum_{k=1}^{n} u_k$ increase monotonically.

Weighted Lebesgue measure. Given $f \geq 0$, define a new set function by

$$\mu(E) \;:=\; \int_E f \, \mathrm{d}m.$$

By MCT, μ is countably additive, hence a measure. It is called a **weighted Lebesgue measure** and written $\mu = f \, \mathrm{d}m$. [4]

[4] Prelude to how every continuous probability measure is defined.

Theorem 5.3: Lebesgue Dominated Convergence Theorem (DCT)

If $f_n \to f$ a.e. and $|f_n| \le g$ for some integrable g, then

$$\lim_{n\to\infty} \int |f_n - f| = 0, \qquad \text{and in particular} \qquad \int f_n \longrightarrow \int f.$$

5.1 Common Counterexamples

Try to avoid the following pathologies when applying convergence theorems.

(i) **Escape to infinity.** $f_n = \frac{1}{n}\chi_{[0,n]} : \mathbb{R} \to \mathbb{R}$. Then $f_n \to 0$ uniformly and $\int f_n = 1$ for all n, but $\int \lim f_n = 0$. *(Mass escapes to $+\infty$.)*

(ii) **Concentrating spike.** $f_n = n \cdot \chi_{[0,1/n]}$. Then $\int f_n = 1$ for all n, but $f_n \to 0$ a.e. *(Mass concentrates at the origin.)*

(iii) **Sliding block.** $f_n = \chi_{[a_n,\, a_n+1/n]}$ for a sequence $a_n \to \infty$. Then $\int f_n = 1/n \to 0$, yet $f_n \to 0$ pointwise. *(Fatou's lemma is tight: $\int f = 0 \le \liminf \int f_n = 0$.)*

6 Convergence in Measure

Definition 6.1: Convergence in Measure

$f_n \to f$ **in measure** if for all $\varepsilon > 0$,

$$m(\{|f_n - f| > \varepsilon\}) \longrightarrow 0 \quad \text{as } n \to \infty.$$

Relation to a.e. convergence.

- $f_n \xrightarrow{\text{a.e.}} f \;\Rightarrow\; f_n \xrightarrow{m} f$ (when $m(E) < \infty$, by Egorov's theorem).
- The converse *fails*: $f_n \xrightarrow{m} f \not\Rightarrow f_n \xrightarrow{\text{a.e.}} f$. *(Exercise: the sliding block above converges in measure but not a.e.)*

Remark 6.1. *Counterexample (iii) above shows $f_n \xrightarrow{a.e.} f$ but $m(\{|f_n - f| > \varepsilon\}) = 1$ does not tend to zero (the sliding block never leaves). However, by Egorov's theorem, if $f_n \to f$ on E with $m(E) < \infty$, one can always find a large subset $A_\varepsilon \subset E$ on which $f_n \to f$ uniformly, giving $f_n \to f$ in measure on E.*

7 Properties of Integrability

Proposition 7.1: Concentration of Integrability[5]

If f is integrable on $\mathbb{R}^d$, then for every $\varepsilon > 0$ there exists $E \subset \mathbb{R}^d$ with $m(E) < \infty$ such that

$$\int_{E^c} |f| \;<\; \varepsilon.$$

Proposition 7.2: Absolute Continuity of the Integral[6]

If f is integrable on $\mathbb{R}^d$, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $E \in \mathcal{M}$ with $m(E) < \delta$,

$$\int_{E} |f| \;<\; \varepsilon.$$

Example (MCT application). For $\alpha > 1$, the Riemann zeta function can be expressed as a Lebesgue integral via MCT:

$$\sum_{n=1}^{\infty} \frac{1}{n^\alpha} = \sum_{n=1}^{\infty} \frac{1}{\Gamma(\alpha)} \int_0^\infty t^{\alpha-1} e^{-nt}\,\mathrm{d}t \overset{\text{MCT}}{=} \frac{1}{\Gamma(\alpha)} \int_0^\infty t^{\alpha-1} \sum_{n=1}^{\infty} e^{-nt}\,\mathrm{d}t = \frac{1}{\Gamma(\alpha)} \int_0^\infty \frac{t^{\alpha-1}}{e^t - 1}\,\mathrm{d}t.$$

(MCT justifies interchanging the sum and integral since all terms are non-negative and partial sums increase.)

[5] Intuitively: integrable functions cannot spread their mass across sets of infinite measure. Proof via MCT: the truncated functions $|f|\chi_{\{|x|<N\}}$ increase monotonically to $|f|$, so their integrals converge to $\int |f|$ by MCT.

[6] Proof via contradiction + Borel–Cantelli + DCT. Assuming no such δ exists produces sets E_n with $m(E_n) < 1/2^n$ and $\int_{E_n} |f| \geq \varepsilon$. By Borel–Cantelli, $m(\limsup E_n) = 0$, so $\int_{\limsup E_n} |f| = 0$. But DCT gives $\int_{\cup_{k\geq n} E_k} |f| \to 0$, contradicting $\int_{E_n} |f| \geq \varepsilon$.

Proofs of Key Results

Theorem 4.3: Chebyshev's Inequality

Proof. By monotonicity of the integral and the fact that $f \geq \alpha$ on $\{f \geq \alpha\}$:

$$\int f \;\geq\; \int_{\{f \geq \alpha\}} f \;\geq\; \alpha \cdot m(\{f \geq \alpha\}).$$

Dividing by $\alpha > 0$ gives the result. □

Lemma 4.4: $\int f < \infty \Rightarrow f < \infty$ a.e.

Proof. For each $n \in \mathbb{N}$, Chebyshev gives $m(\{f \geq n\}) \leq \frac{1}{n} \int f$. The sets $\{f \geq n\} \searrow \{f = +\infty\}$; by downward continuity of measure,

$$m(\{f = +\infty\}) = \lim_{n\to\infty} m(\{f \geq n\}) \leq \lim_{n\to\infty} \frac{1}{n} \int f = 0.$$

□

Lemma 4.5: $\int f = 0 \Rightarrow f = 0$ a.e.

Proof. For each $n \in \mathbb{N}$, Chebyshev gives $m(\{f \geq 1/n\}) \leq n \int f = 0$. The sets $\{f \geq 1/n\} \nearrow \{f > 0\}$; by upward continuity of measure, $m(\{f > 0\}) = \lim_{n\to\infty} m(\{f \geq 1/n\}) = 0$. □

Theorem 2.4: Bounded Convergence Theorem

Proof. Let $M := \sup_E |f| < \infty$. Fix $\varepsilon > 0$; by Egorov's theorem, find $A_\varepsilon \subset E$ with $m(E \setminus A_\varepsilon) < \varepsilon$ on which $\varphi_j \to f$ uniformly.

Part (i): $\lim_j \int_E \varphi_j$ exists. Set $I_j := \int_E \varphi_j$ and show $\{I_j\}$ is Cauchy. For any $n, m \in \mathbb{N}$:

$$|I_n - I_m| \leq \int_E |\varphi_n - \varphi_m| = \int_{A_\varepsilon} |\varphi_n - \varphi_m| + \int_{E\setminus A_\varepsilon} |\varphi_n - \varphi_m| \leq m(E) \cdot \sup_{A_\varepsilon} |\varphi_n - \varphi_m| + 2M \cdot \varepsilon.$$

Since $\varphi_j \to f$ uniformly on A_ε, the sequence is uniformly Cauchy there, so the first term $\to 0$ as $n, m \to \infty$. Thus $\{I_j\}$ is Cauchy in $\mathbb{R}$, hence convergent.

Part (ii): $f = 0$ a.e. $\Rightarrow \lim_j \int_E \varphi_j = 0$. By Egorov, $\varphi_j \to 0$ uniformly on A_ε:

$$\left| \int_E \varphi_j \right| \leq \int_{A_\varepsilon} |\varphi_j| + \int_{E\setminus A_\varepsilon} |\varphi_j| \leq m(E) \cdot \sup_{A_\varepsilon} |\varphi_j| + M \cdot \varepsilon \;\longrightarrow\; M\varepsilon \quad \text{as } j \to \infty.$$

Since $\varepsilon > 0$ is arbitrary, $\lim_j \int_E \varphi_j = 0$. □

Theorem 5.1: Fatou's Lemma

Proof. WLOG assume $f_n \to f$ pointwise everywhere (remove the a.e. null set at no cost). Let g be any bounded function with $m(\text{supp}(g)) < \infty$ and $0 \le g \le f$. Set $g_n := \min\{g, f_n\}$; then $g_n \to g$ pointwise, $|g_n| \le M := \sup g$, and $\text{supp}(g_n) \subset \text{supp}(g)$.

By the Bounded Convergence Theorem, $\int g_n \to \int g$. By monotonicity, $g_n \le f_n$ so $\int g_n \le \int f_n$. Taking the limit inferior:

$$\int g = \lim_n \int g_n = \liminf_n \int g_n \le \liminf_n \int f_n.$$

Since this holds for all such $g \le f$, taking the supremum over g and using the definition of $\int f$ for positive functions:

$$\int f \;\le\; \liminf_{n\to\infty} \int f_n.$$

□

Theorem 5.2: Monotone Convergence Theorem

Proof. Since $f_k \nearrow f$, monotonicity of the integral gives $\int f_k \le \int f$ for all k, and the sequence $\{\int f_k\}$ is itself non-decreasing. Thus $\lim_k \int f_k \le \int f$.

The reverse inequality follows from Fatou's Lemma:

$$\int f \;\le\; \liminf_{k\to\infty} \int f_k = \lim_{k\to\infty} \int f_k.$$

□

Theorem 5.3: Dominated Convergence Theorem

Proof. Since $|f_n| \le g$ and $f_n \to f$ a.e., we have $|f| \le g$, and $|f_n - f| \le |f_n| + |f| \le 2g$. Thus $2g - |f_n - f| \ge 0$.

Apply Fatou's Lemma to $2g - |f_n - f| \ge 0$ (using $f_n \to f$ a.e.):

$$\int 2g \;\le\; \liminf_{n\to\infty} \int \big(2g - |f_n - f|\big) = \int 2g - \limsup_{n\to\infty} \int |f_n - f|.$$

Subtracting $\int 2g < \infty$ from both sides:

$$0 \;\le\; -\limsup_{n\to\infty} \int |f_n - f|, \quad \text{i.e.} \quad \limsup_{n\to\infty} \int |f_n - f| \;\le\; 0.$$

Since $\int |f_n - f| \ge 0$, we conclude $\lim_{n\to\infty} \int |f_n - f| = 0$. □

Proposition 7.1: Concentration of Integrability

Proof. Define $E_N := \{|x| < N\}$. The functions $|f|\chi_{E_N}$ are measurable and increase pointwise to $|f|$. By MCT:

$$\lim_{N\to\infty} \int |f|\chi_{E_N} = \int |f| < \infty.$$

Hence there exists N such that $\int_{E_N^c} |f| = \int |f| - \int_{E_N} |f| < \varepsilon$. Take $E = E_N$. $\square$

Proposition 7.2: Absolute Continuity of the Integral

Proof. Suppose for contradiction no such δ exists. Then for some $\varepsilon > 0$ and every $n \in \mathbb{N}$, there is a measurable set E_n with $m(E_n) < 1/2^n$ yet $\int_{E_n} |f| \geq \varepsilon$.

Since $\sum_n m(E_n) < \infty$, the Borel–Cantelli Lemma gives $m(\limsup_n E_n) = 0$, so $\int_{\limsup E_n} |f| = 0$.

The functions $|f|\chi_{\bigcup_{k\geq n} E_k}$ are dominated by the integrable function $|f|$ and decrease to $|f|\chi_{\limsup E_n}$ pointwise. By DCT:

$$\lim_{n\to\infty} \int_{\bigcup_{k\geq n} E_k} |f| = \int_{\limsup E_n} |f| = 0.$$

But $E_n \subset \bigcup_{k\geq n} E_k$, so $\int_{E_n} |f| \leq \int_{\bigcup_{k\geq n} E_k} |f| \to 0$, contradicting $\int_{E_n} |f| \geq \varepsilon$. $\square$